MUHAMMAD ZAMZURI BIN ZAMANI

Electronics Engineering Student (Analog & RF IC Design) ◇ Nationality: Malaysian

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OBJECTIVE

Seeking a **12–15 week internship starting 24 May 2027** in **analog / RF IC design**, to design and verify **RF analog building blocks** (oscillators/PLLs, LNAs, data converters) with hands-on CMOS design in **Cadence Virtuoso** and **Python/MATLAB** scripting.

EDUCATION

Master of Engineering (Diplôme d'Ingénieur) in Electronics **2024 - 2028**
Enseirb-Matmeca, Bordeaux INP *Talence, FR*

- **Analog/RF courses:** Integrated Circuits, **Feedback Amplifiers**, **Oscillators**, **Voltage References**, Analog & Digital Modulations, Signal Conditioning
- **Digital courses: VHDL Synthesis & FPGA Design**, Digital Circuit Technologies, 32-bit MCUs, DSP

Bachelor (B.U.T.) of Electrical Engineering and Industrial Computing **2022 - 2024**
*IUT Nice Côte d'Azur, Université Côte d'Azur — Electrical and Energy Management, equivalent **3.5 CGPA*** *Nice, FR*

Foundation in Engineering (French Engineering Preparatory Programme) **2021 - 2022**
UniKL MFI (Malaysia France Institute) *Kuala Lumpur, MY*

ANALOG / RF IC DESIGN PROJECTS

Negative-Resistance LC Oscillator in 350 nm CMOS **2026**

- Designed a **cross-coupled NMOS** oscillator whose **negative resistance** ($-2/g_m$) cancels the losses of two 10 nH ($Q_L = 20$) inductors with 6.76 pF ($R_T \approx 544\,\Omega$): **433 MHz**, **1.75 V** differential amplitude at **2.2 mA** from 3.3 V, **−118.8 dBc/Hz at 100 kHz** (spec < -110).
- Sized devices by hand ($g_m = 2\times$ start-up value); verified tank (S-param), start-up, amplitude and **phase noise** with **tran, PSS and pnoise** in **Cadence Virtuoso (Spectre)**.

4-bit Flash ADC in AMS 0.35 μm BiCMOS (team of 2) **2025**

- Designed bottom-up in **Cadence Virtuoso**: transmission-gate **sample-and-hold**, current-mirror biasing, N/P **comparators**, 16-level ladder (**206 mV LSB**), decoder and register; full ADC verified in transient.

Optoelectronic FSK Transmission Link — Complete TX-to-RX Chain **2026**

- Designed the full optical link from **FSK modulation** and LED driving (TX) to photodiode **amplification, band-pass filtering** and **envelope-detection demodulation** (RX): **1200 baud** over 30 cm from a single 5 V rail.

4-bit R-2R DAC with Sample-and-Hold Deglitching (SPICE) **2026**

- R-2R ladder DAC (**312.5 mV LSB**) with op-amp buffer; a synchronised **sample-and-hold** (20 μs blind-spot clock) removed a 4.73 V **major-carry glitch**, and a slower track RC (τ: 1→10 μs) cancelled a **12 mV overshoot**.

RESEARCH EXPERIENCE

R&D Engineer - INTERNSHIP **Jun 2026 - Aug 2026**
Graduate Student Researcher / R&D Intern **Jun 2025 - Aug 2025**
LDMRC, Department of Physics, Universiti Malaya *Kuala Lumpur, MY*

- Wrote **Python scripts** to generate **interdigitated electrode (IDE)** geometries, routed them in **KiCad**, fabricated them and deposited PEDOT:PSS films for capacitive humidity sensors (**co-authored publication**).

R&D Hardware/Software - INTERNSHIP **Feb 2024 - Apr 2024**
LEAT, CNRS – Université Côte d'Azur *Biot, FR*

- Designed the control board and **STM32** C firmware switching a reconfigurable **PIFA antenna** between bands.

PUBLICATION

M. Qianhui, M. Alserhan, **Z. Zamani**, *et al.*, "Humidity-induced capacitance as a probe of formulation-specific phase separation in PEDOT:PSS thin films," *Surfaces and Interfaces*, vol. 98, art. 110567, 2026. DOI: 10.1016/j.surfin.2026.110567

SKILLS

Analog / RF	CMOS transistor sizing, LC oscillators, phase noise, ADC/DAC, optical TX/RX chains
EDA Tools	Cadence Virtuoso (Spectre: AC, tran, S-param, PSS, pnoise), SPICE, ADS, MATLAB, KiCad
Scripting / Digital	Python, MATLAB, C, C++; VHDL, Verilog, Vivado, ModelSim, Quartus; STM32, ESP32, PIC
Languages	Malay (**Native**), English (**Bilingual**), French (**Fluent**), Thai (**Basic**), Mandarin (**Basic**)

LEADERSHIP

- **Project Manager**, Malaysian Students' Association in France (2023–26): led the **"From Strategy to Silicon"** semiconductor forum (**100+** students).